Experiment No. 10

Aim : Implementation and Presentation of Mini Project.

Title : Legal Contract Clause Extractor (LCCE)

Legal Contract ClauseExtractor (LCCE)

Automated Contract Intelligence using NLP



Developed & Presented by:

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Experiment 10:

Implementation & Demonstration

Problem Statement & Motivation

THE PROBLEM

Modern legal contracts are dense, jargon-heavy, and complex

Often hide predatory terms like extreme liability caps, auto-renewal traps, and unilateral fee shifting

THE MOTIVATION

Legal counsel costs ~$300+/hr, unaffordable for most

Leads to significant power imbalances in contract negotiations

THE GOAL

Build an intelligent NLP web app as a free "AI Co-Counsel."

Democratize legal comprehension by translating dense legalese into plain English

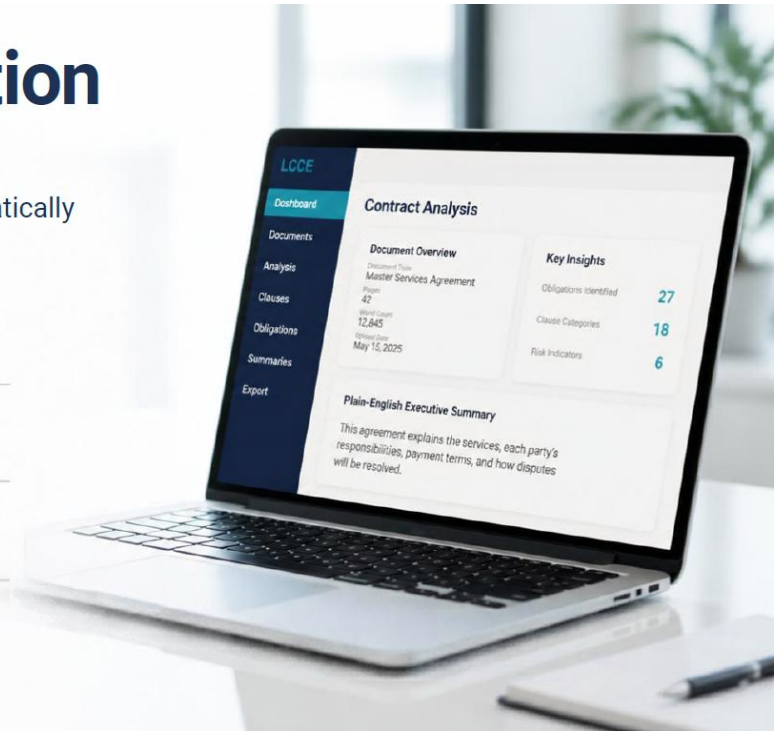


Proposed Solution

THE LCCE PLATFORM

A full-stack web application that automatically parses and analyzes legal documents.

- Seamless ingestion of raw text or PDF documents
- Advanced generative AI delivers semantic contract understanding
- Extracts key metadata, identifies binding obligations, and categorizes clauses
- Creates plain-English executive summaries written at a 10-year-old level



Technology Stack

FRONTEND GUI
HTML5, Advanced CSS3
OLED Dark Mode, Micro-animations
Vanilla JavaScript, Chart.js

BACKEND API
Python 3.11
Flask WSGI framework

NLP / AI PIPELINE
OpenAI Python SDK & OpenRouter API Gateway
Google Gemma (26B)
Strict ISO-Schema Prompt Engineering

EXPORT TOOLING
xhtml2pdf for converting
web intelligence into A4 PDF reports



NLP Algorithm & System Architecture



1. **User uploads PDF; Flask backend extracts raw text, preserving structure**
2. **Backend creates strict Pydantic JSON Schema and injects it into the AI prompt**
3. **Google Gemma processes text via OpenRouter; exponential backoff handles API rate limits**
4. **Raw JSON from response is sanitized, parsed, then sent to the frontend**
5. **JavaScript updates the DOM, renders canvas charts, and highlights critical risks dynamically**

Key Features

INTELLIGENCE DASHBOARD

Instant risk score and multi-axis analysis across key contract dimensions

RISK MITIGATION ENGINE

Pinpoints predatory clauses and delivers actionable strategies to strengthen terms

DYNAMIC PDF REPORTING

Export insights, visuals, and plain-English summaries in a structured, downloadable PDF



Project Demonstration

(Part 1 – UI & Analysis)

Live walkthrough of the LCCE web application

Upload and parse contracts in seconds

Real-time NLP analysis with interactive visualizations

Color-coded risk alerts to assess contract exposure



Project Demonstration

(Part 2 - Output & Conclusion)

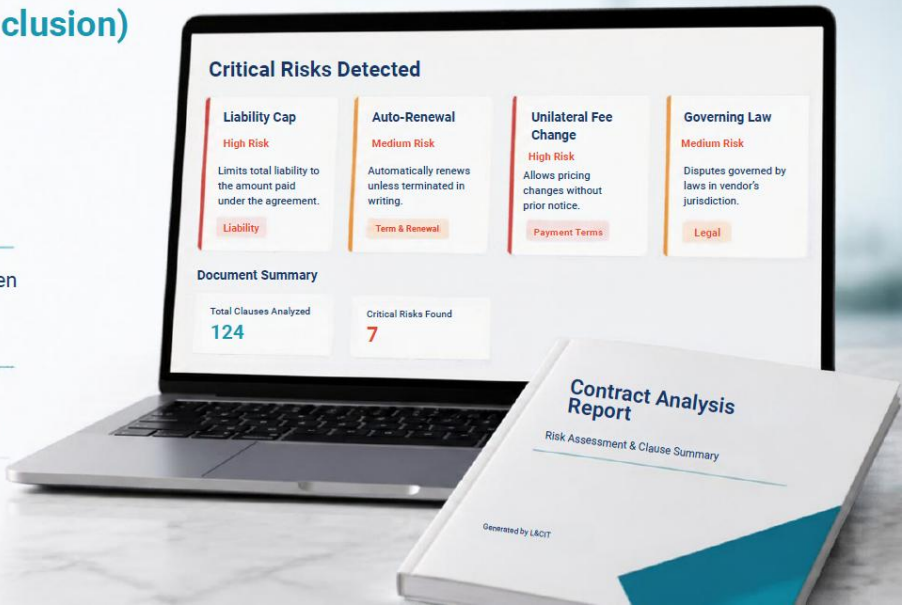
PROJECT SUMMARY

LCCE proves Generative NLP can automate complex legal review

Expert clause categorization and hidden risk extraction empower users with negotiation leverage

Seamless PDF export makes intelligence portable and professional

Fulfills the goal of democratizing legal comprehension effectively



Thank You...

Conclusion : Hence we successfully completed the project & presented it.